THE NEED FOR SPEED: ULTRAFAST BREAST MRI FOR SCREENING

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BACKGROUND

Breast cancer remains a leading cause of mortality among Malaysian women, creating an urgent demand for rapid, accessible screening solutions. Standard MRI is hindered by long scan times and high costs, but Ultrafast Breast MRI revolutionizes screening by capturing early microvascular kinetics within seconds post-contrast..

OBJECTIVES

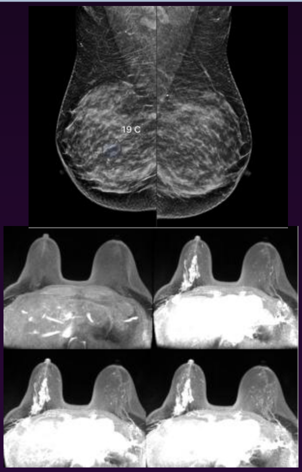
To revolutionize breast screening by establishing high-temporal-resolution Ultrafast MRI as an ultra-rapid, highly specific adjunct for definitive lesion characterization across diverse clinical cohorts.

METHODS

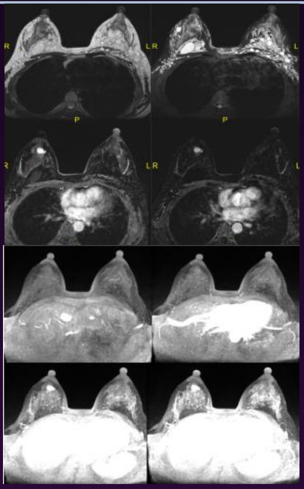
A retrospective review was conducted on 109 patients who underwent 3T contrast-enhanced breast MRI at UMMC, including high-risk BRCA screening, pre-operative assessment for newly diagnosed breast cancer, and evaluation of patients with breast filler injections. The study utilized an integrated protocol adding high-temporal ultrafast sequences (2–10 s/frame) as a dynamic adjunct to standard MRI. The **Maximum Slope (MS)** of early enhancement from the ultrafast protocol was evaluated as the key quantitative parameter for direct comparison against the standard kinetic curve of conventional breast MRI.

RESULTS

Real-World Case Demonstrations: Ultrafast Breast MRI



Case 1 (Invasive Carcinoma In Situ): A 59-year-old female with bloody nipple discharge and dense breasts showed a spiculated mass with segmental non-mass enhancement on Ultrafast MRI. Histopathology confirmed invasive carcinoma in situ.



Case 2 (Fibroadenoma): A 42-year-old female with breast filler injections presenting with a palpable lump demonstrated a lesion with characteristically slower early enhancement kinetics. Histopathology confirmed a benign fibroadenoma.

Key numerical findings

Kinetic Performance: Malignant lesions demonstrated a significantly steeper Maximum Slope compared to benign lesions (13.27%/s vs 5.45%/s, $p<0.0001$).
Diagnostic Metrics: Ultrafast MRI achieved superior specificity (71.43% vs. 59.09%) compared to standard MRI, while maintaining comparable overall diagnostic accuracy (80.00% vs. 81.25%).
Clinical Yield: Early vascularity capture substantially reduced false positives, minimizing unnecessary follow-ups and invasive biopsies

CONCLUSION / TAKE-HOME MESSAGE

1. Main finding

Ultrafast MRI delivers a sharp gain in diagnostic specificity (71.43% vs. 59.09% for standard MRI) via early Maximum Slope analysis, while matching conventional overall diagnostic accuracy (80.00% vs. 81.25%).

2. Clinical meaning

This game-changing adjunct protocol fundamentally shifts the diagnostic paradigm—capturing tumor angiogenesis within seconds to dismantle false-positive rates, spare women from unnecessary invasive biopsies, and streamline clinical decision-making.

3. Next step

Integrate machine learning algorithms to automate Maximum Slope analysis and validate standardized cutoff values in multi-center clinical trials for standalone abbreviated MRI protocols

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